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NSD, PHILA., PA.
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MILITARY SPECIFICATION

DETECTING SET, SUBMARINE: AN/ASR-3

This specification has been approved by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide signals which indicate a change in the number of condensation nuclei present in the air entering the equipment. It has been designed to be carried by aircraft for the purpose of detecting exhaust trails from submarines either running on the surface or snorkeling.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size Inches	Max. Allow. weight Lbs.	Applicable Paragraphs
Detector, Particle	DT-193/ASR-3	MS91403-B1D1	38	3.3.1
Control, Detecting Set	C-2293/ASR-3	5-3/4 6 4	3	3.3.2
Meter, Arbitrary Scale	ME-130/ASR-3	4(DIA) 2-3/4	1-1/2	3.3
Mounting Base	MT-B1D1	MS91405-B1D1	3	3.3.1.2

1.3 Accessories - The equipment shall operate with the following accessories. However, these accessories are not to be supplied as part of this equipment.

Recorder

Cables

Scoop

Air Tube

Intercommunications System

Dead Reckoning Tracer (DRT)

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

2.2 SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes, Choice and Application of
MIL-C-172	Cases: Bases, Mounting: and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-I-6181	Interference Control Requirements, Aeronautical Equipment
MIL-P-7788	Plate, Plastic. Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft, Characteristics of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-P-17555	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical and Electromechanical)
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection Format for
MIL-A-6752	Ammeters, Voltmeter, and Loadmeters, Direct Current

Bureau of Aeronautics

SAR-300	Change Proposals, Material Changes and Bulletins, Preparation of

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STANDARDS

Military

MS-25245	Connections - Input Power Airborne Electronic Equipment, 28 Volts DC-115 Volts Single Phase AC, and/or 115-200 Volts 3 phase AC

2.3 Availability of Documents -

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot (CDS) Philadelphia 11, Pennsylvania.

2.4 Precedence - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the Procuring Agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of Parts and Materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and Materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet the requirements of the respective standard specification.

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(3) When a non-standard part is used, the meeting of general material requirement such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall first be approved by the procuring agency.

3.1.1 Non-Standard Approval and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in Specification MIL-E-5400 of an electrical nature must be approved before the preproduction model of the equipment is delivered to the Navy. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the NON-Standard Part Data Sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Performance Objectives - Minimum size and weight, simplicity of operation and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives to be considered in the production of the equipment specified herein. The use of materials, parts and processes other than those required by Specification MIL-E-5400 shall be investigated and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their use, a request for approval of a non-standard part shall be submitted to the procuring agency for consideration, each request should be accompanied by complete supporting information.

3.1.3 Tube Types - The choice and application of electron tubes used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

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(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.4 Electron Tube Complement - A tube complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.

3.1.5 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of MIL-E-5400.

3.2 General Requirements -

3.2.1 General - The equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance and for making laboratory bench test on the equipment:

Temperature	Room ambient 25°C (+15°C or -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power Voltage	115 ± 1V/ 200 ± 2V, 400 ± 20 cps, three-phase a-c, four-wire, Y-connected and 27.5 ± 0.5 V d-c

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in MIL-E-5400 (Class I) with the exception of the conditions noted in the following paragraphs.

3.2.3.1 Operating Altitude - The equipment shall be required to operate satisfactorily at altitudes up to 18,000 feet; however, the equipment shall not be damaged when operated at altitudes up to 50,000 feet nor shall the equipment become an explosion or fire hazard when operated at altitudes up to 50,000 feet.

3.2.4 Total Weight - The total weight of the equipment excluding cables and water shall not exceed 45-1/2 pounds.

3.2.5 Operational Stability - The equipment shall operate within equipment specification for at least 50 hours, continuously or intermittently, without the necessity for readjustment of any controls which are nonaccessible during flight. A minimum of ten hours operation is required without replenishing the water in the equipment under operating ambient conditions of up to 90° F temperature, and with relative humidity as low as 30%.

3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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3.2.7 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of specification MIL-W-5088. A minimum number of individually shielded wires shall be used. Shielding shall be used when demonstrated as necessary to meet the interference control requirements of MIL-I-6181.

3.2.8 Cables and Connectors - The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the Bureau of Aeronautics prior to preproduction testing.

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

Operating Voltage Limits	102 to 124 volts
Operating Frequency Limits	380 to 420 cps
Power (3Ø)	500 VA

3.2.11 Input Power Connection - Input power connections shall be in accordance with Drawing MS-25245.

3.2.12 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.12.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Moisture and Fungus Resistant Treatment - Equipment shall be fungus proofed by selection of parts and materials as non-nutrient for fungus, or the parts and materials shall be treated prior to their use in the equipment so that over-all spraying of the equipment is not necessary.

3.2.13.1 Demineralized Water - Demineralized water shall be used in the humidity chamber whenever sediment formation is likely due to direct use of tap water.

3.2.14 Nomenclature and Nameplates - Nomenclature assignment and Nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.15 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units of the equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for that purpose. In the event that requirements of Specification MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

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3.2.16 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of Specification MIL-E-5400.

3.2.17 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.18 Performance -

3.2.18.1 Warm-Up Time - The warm-up time shall not exceed one minute when the humidifier water is not frozen and shall not exceed 30 minutes when the ambient temperature is 0°C to -55°C.

3.2.18.2 Sampling Rate - The sampling rate shall be not less than 5 per second.

3.2.18.3 Differential Pressure - The differential pressure in the expansion chamber shall be adjustable over at least the range of 4 to 12 inches of mercury vacuum. The average differential pressure shall be automatically maintained within ± 1/4 inch of mercury at any preselected setting.

3.2.18.4 Humidity Content - A humidity chamber shall be provided containing an adequate method for properly humidifying the sampled air.

3.2.18.5 Sensitivity - The equipment shall be capable of rendering an indication on as little as 50 nuclei/cc. The determination of the 50-nuclei/cc level may be done by extrapolating the indicating instrument reading from a level no higher than 250 nuclei/cc.

3.2.18.6 Sensitivity Control - The sensitivity shall be controlled by a manually adjustable control located on the Detecting Set Control. The positions of the control shall be marked to indicate relative sensitivity.

3.2.18.7 Indications -

(1) The signal shall be presented on one of the pens of the recorder.

(2) The same signal shall be displayed on the function meter on the detecting set control.

(3) The same signal shall be displayed on a remote panel meter to be located in the pilot's compartment.

(4) Provision shall be made for a signal indication to be aurally presented.

(5) Provision shall be made for a signal indication to be presented to a Dead Reckoning Tracer (DRT).

3.2.18.8 Compensation - The equipment shall include, if necessary, compensation in the photomultiplier light source circuit so designed that input voltage variations within the limits specified in paragraph 3.2.10 shall not cause fluctuations in output indication greater than 5%.

3.2.18.9 Vacuum - The vacuum shall be capable of being preset by an adjustment accessible from the front panel of the particle detector.

3.3 Detail Requirements -

3.3.1 Particle Detector, DT-193/ASR-3 - This unit shall meet the following requirements.

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3.3.1.1 Function -

(1) The particle detector shall perform the electrical and mechanical function of converting the condensation nuclei entering the unit into electrical energy suitable for display as indications of signal.

(2) The particle detector shall provide the proper voltage for the operation of the unit.

(3) The particle detector shall provide output signals for indicators.

3.3.1.2 Form Factor and Mounting - The particle detector shall be enclosed in a size B1D1 case and mounted on a vibration isolator type mounting, size MT-B1D1. Case and mounting shall meet the applicable requirements of Specification MIL-C-172. Electrical connectors shall be provided at the rear of the unit only.

3.3.1.3 Weight - The weight of this unit, including all components and mounting provisions shall not exceed thirty-eight pounds.

3.3.1.4 Contents of Unit - The particle detector shall contain the following subassemblies:

3.3.1.4.1 Amplifier Subassembly - The amplifier subassembly shall fulfill the following requirements:

(1) The amplifier subassembly shall provide a voltage sufficient to electrically offset the RD-47A/ASQ-8 recorder pen 25 ± 2.5 scale divisions to the left of center with zero input signal, and to provide a signal to drive the recorder pen 50 ± 2.5 divisions to the right of the left margin when the atmosphere entering the expansion chamber contains 500 nuclei per cubic centimeter as measured on a Nolan photo-electric nucleus counter or approved equivalent.

(2) The amplifier subassembly shall provide a marking signal for the DRT to be initiated in conjunction with the ICS alarm signal. The marking signal shall consist of a 400 cps, 3 phase half wave rectified sine wave, the average value of which is 28 volts d-c when connected to a 680 ohm resistive load.

(3) The amplifier subassembly shall provide a 1-1/4 second ± 1/2 second audio tone of 1200 ± 200 cps for injection into the aircraft ICS system. The particle detector (generator) shall supply 250 milliwatts to a load of 600 ohms. The tone shall be initiated by an instantaneous increase in signal level. The alarm sensitivity shall be preadjustable so that a predetermined increase in signal level (to any point within the range of 10 per cent to 50 per cent of full scale deflection on the particle level meters) will initiate the alarm tone.

(4) The amplifier subassembly shall be capable of being removed by quick disconnect from an internal plug assembly.

(5) All controls required for flight operation of the amplifier subassembly shall be located on the detecting set control unit.

(6) Pin jacks shall be provided on the front panel of the particle detector unit to duplicate the signal voltage to the particle level meters.

(7) The amplifier subassembly shall provide an output sufficient to deflect two - 1 ma, 1,000 ohm meters to full scale under the same conditions as (1) above.

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(8) The dynamic response characteristics of the amplifier shall be essentially square law and shall fall within the envelope shown in Figure 1.

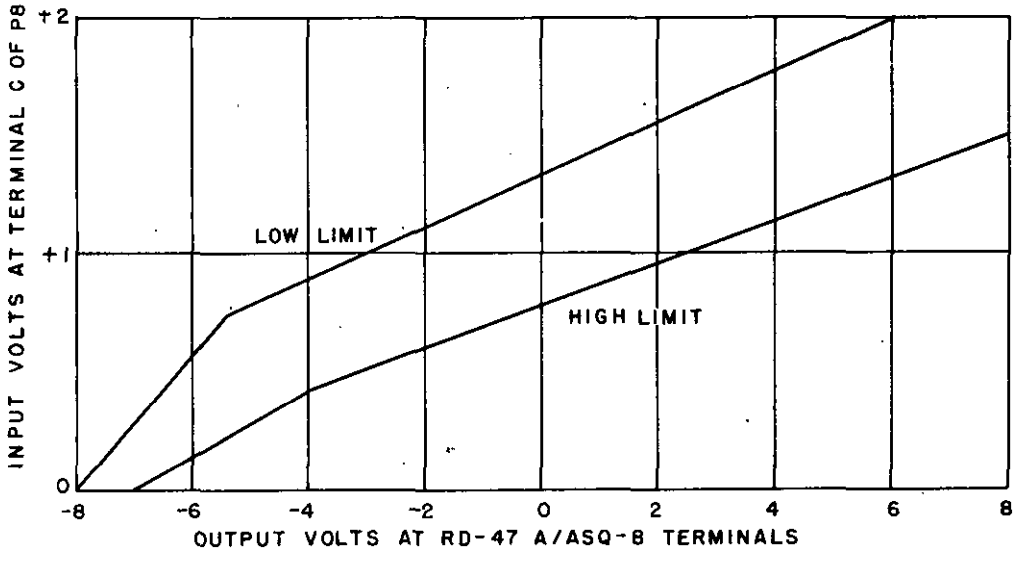


FIGURE 1. DYNAMIC RESPONSE CHARACTERISTICS

FIGURE 1

3.3.1.4.2 Detector Subassembly - The detector subassembly shall fulfill the following requirements:

(1) The detector subassembly shall be an integral part of the main chassis and shall be accessible for minor servicing by removal of the particle detector dust covers.

(2) The detector shall have provision on the front panel for connecting the input air under observation.

(3) The detector shall be designed so that the vacuum pump cannot drain water from the humidifier when the unit is tilted in any direction through angles up to 45 degrees from the horizontal.

(4) The detector expansion shall be accomplished by a vacuum pump which will derive its power from 115/200 V, 400 cycle three phase, and shall be capable of pulling 15 inches of mercury at sea level.

(5) The detector control valve shall maintain constant pressure differential across the expansion chamber. This constant differential pressure shall be capable of being adjusted and locked conveniently by means of an adjustment on the front panel.

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(6) A humidifier for properly conditioning the air sampled shall be provided.

(7) The detector shall incorporate an optical system, light source and phototube arranged to permit measurement of the light scattered by condensation within the expansion chamber. In the absence of condensation nuclei in the air being sampled, the stray light impinging on the phototube shall not limit the sensitivity to below that specified in paragraph 3.2.13.5.

(8) The detector shall contain a motor driven rotary valve to cycle the air samples. This rotary valve shall be capable of flushing, filling, dwelling, and expanding the air sample at a rate of no less than 5 cycles per second.

(9) Heaters and a thermostat shall provide safe operation of the equipment during freezing weather. These heaters shall operate automatically and provision shall be made to power an indicating light which may be remotely located. An automatic control shall be provided so that voltage cannot be applied to the motors until all moving parts are capable of normal operation after having been exposed to freezing temperature.

(10) A method of adding water to the humidifier shall be provided for on the front panel. A sight gauge shall be provided to indicate proper operational limits of the water level on the front panel.

(11) A facility to attach a standard gauge for measuring the differential pressure across the expansion chamber shall be provided for on the front panel.

(12) The water level gauge shall be illuminated while the equipment is in operation.

3.3.1.4.3 Power Supply Subassembly(s) - The power supply subassembly shall fulfill the following requirements:

(1) The power supply subassembly has been designed to convert 400 cycle, three-phase 115/200 V a-c into the necessary d-c plate supply and the appropriate a-c voltages for filament and indicator light demands.

(2) The power supply subassembly shall be capable of being readily removed.

(3) The power supply unit shall provide regulation of the required voltages such that input voltage variations as specified in paragraph 3.2.10 shall not cause any fluctuations in output indications greater than 5%.

3.3.1.4.4 Power and Distribution Connectors - Power and distribution connectors shall be located in the rear of the particle detector. The particle detector shall also contain integral connectors to facilitate the connection of the amplifier and power supply subassemblies which will be plugged in by suitable arrangement within the particle detector.

3.3.2 Detecting Set Control C-2293/ASR-3 - This unit shall meet the following requirements:

3.3.2.1 Function - The unit shall provide for remote control of the particle detector unit.

3.3.2.2 Form Factor and Mounting - The unit shall be a rack mounted console control conforming to the applicable requirements of Specification MIL-C-6781, having dimensions not to exceed 5-3/4 inches wide, 6 inches long, and 4 inches deep.

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3.3.2.3 Content of Unit - The following controls shall be mounted on a lighted plastic plate:

(1) GAIN Control

(2) PARTICLE LEVEL Meter

(3) POWER (ON-OFF) Switch

(4) "On-Off" Indicator light for humidity chamber heater

3.3.3 Arbitrary Scale Meter ME-130/ASR-3 - The arbitrary scale meter shall fulfill the following requirements:

(1) The unit shall be a 0-1 volt d-c full scale, 1000 ohm/volt ± 3% meter.

(2) The unit shall be suitable for mounting on the pilot's console panel.

(3) The meter shall meet the requirements of specification MIL-A-6752 except as specified herein.

(4) The meter shall be mounted by means of four screws equally spaced on a 3.500 ± 0.005 inch diameter bolt circle.

(5) Scale markings and graduations shall be submitted for approval to the Bureau of Aeronautics.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodially through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minute cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol-number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

6.3 The parentheses (*), when used, in the type designation will be replaced by either a number or letter furnished by the Bureau of Aeronautics upon application by the contractor for assignment of nomenclature in accordance with 3.2.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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